

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Cilium Assembly (GO:0060271)	-0.07760309	197	1.877e-04	5.039e-01	CEP126:1 DYNC2H1:5 KIAA0753:28 GORAB:36 TEK1:92 DNAH5:138
Replication Fork Processing (GO:00031297)	-0.18268529	37	1.220e-04	5.039e-01	EME2:124 BRCA1:206 EME1:237 MUS81:419 RAD50:740 DNA2:775
DNA Repair (GO:0006281)	-0.06268710	234	1.031e-03	9.225e-01	USP45:10 XRC5:31 ATM:60 MLH1:66 MLH3:90 EME2:124
G Protein–Coupled Glutamate Receptor Sig	0.32713099	9	6.790e-04	9.225e-01	TRPM1:87 HOMER3:113 GRM5:318 GRM8:415 GRM1:445 GRM6:491
Double–Strand Break Repair (GO:0006302)	-0.08361591	134	8.687e-04	9.225e-01	ZMYND8:23 XRC5:31 ATM:60 EME2:124 RNF169:132 BRCA1:206
Positive Regulation Of Cellular Component	0.11602502	70	8.064e-04	9.225e-01	CLU:5 TGFBI:22 ATP10A:56 LRP1:67 BID:143 DCTN1:267
'De Novo' AMP Biosynthetic Process (GO:0)	-0.10120681	4	4.833e-01	9.602e-01	PFAS:33 PAICS:6785 ATIC:6785 ADS51:6785 NA NA
'De Novo' Post–Translational Protein Fold	0.12126361	15	1.040e-01	9.602e-01	SDF2L1:76 GAK:122 HSPA8:475 ENTDP5:5881 PTGES3:5881 DNAJB13:5881
2–Oxoglutarate Metabolic Process (GO:000)	0.05634844	13	4.819e-01	9.602e-01	OGDHL:151 D2HGDH:960 IDH1:5881 G0T2:5881 G0T1:5881 IDH2:5881
3–UTR–mediated mRNA Destabilization (GO:0)	-0.22184976	11	4.171e-01	9.602e-01	MOV10:609 DHX36:916 RC3H1:5881 ZFP36L1:5881 ZFP36L2:5881 ZFP36:5881
3–UTR–mediated mRNA Stabilization (GO:0)	-0.06115460	11	2.229e-01	9.602e-01	DAZL:107 MAPK14:886 BOLL:963 ZFP36:5881 TARDBP:5881 YB3:5881
3–Phosphoadenosine 5–Phosphosulfate Me	-0.04132683	8	6.943e-01	9.602e-01	ENPP1:436 PAPSS2:1069 TPST2:6785 SULT2B1:6785 SULT1B1:6785 SULT1E1:6785
5S Class rRNA Transcription By RNA Polym	-0.05186503	5	6.880e-01	9.602e-01	GTF3C5:1509 GTF3C2:6785 GTF3C3:6785 GTF3CA:6785 GTF3C6:6785 NA
7–Methylguanosine RNA Capping (GO:000945)	0.25805223	2	2.062e-01	9.602e-01	RNMT:188 CMTR2:5881 NA NA NA NA
7–Methylguanosine mRNA Capping (GO:00063)	0.25805223	2	2.062e-01	9.602e-01	RNMT:188 CMTR2:5881 NA NA NA NA
ADP Transport (GO:0015866)	0.11126131	5	3.889e-01	9.602e-01	SLC25A41:826 SLC25A42:5881 SLC25A24:5881 SLC25A5:5881 SLC25A23:5881 NA
AMP Biosynthetic Process (GO:0006167)	-0.13442773	6	2.542e-01	9.602e-01	PFAS:33 APTT:864 PAICS:6785 ATIC:6785 ADS51:6785 NUDT2:6785
AMP Metabolic Process (GO:0046033)	0.07641078	9	4.274e-01	9.602e-01	NT5E:365 AK9:1217 ADS51:5881 NUDT2:5881 NT5C1A:5881 AMPD1:5881
ATF6–mediated Unfolded Protein Response	-0.05345055	5	6.789e-01	9.602e-01	DDIT3:1400 MBTFS1:6785 XBP1:6785 ATF6:6785 CREBZF:6785 NA
ATP Metabolic Process (GO:0046034)	0.03454527	25	5.501e-01	9.602e-01	TGFB1:22 MYH7:163 NT5E:365 HSPA8:475 NUDT2:5881 ATP5PO:5881
ATP Synthesis Coupled Electron Transport	-0.11555397	3	4.882e-01	9.602e-01	NDUFV3:1179 NDUFB6:6785 NDUF52:6785 NA NA NA
ATP Transport (GO:0015867)	0.02195318	9	8.196e-01	9.602e-01	SLC25A41:826 SLC25A42:5881 SLC25A24:5881 SLC25A5:5881 SLC25A23:5881 CALHM3:5881
AV Node Cell Action Potential (GO:008601)	0.10753371	6	3.617e-01	9.602e-01	SCN10A:124 CACNB2:5881 CXNA5:5881 SCN4B:5881 CACNA1G:5881 TRPM4:5881
AV Node Cell To Bundle Of His Cell Commu	0.03089431	4	8.306e-01	9.602e-01	SCN5A:5881 CXNA5:5881 GC3H:5881 GJA5:5881 NA NA
Arp2/3 Complex–Mediated Actin Nucleation	-0.19273439	10	3.485e-02	9.602e-01	WASHC1:500 ARP5C:662 WHAMM:1033 ARP5C1:1228 TRIM27:1940 ARPCL1A:2019
B Cell Activation (GO:0042113)	0.05020335	61	1.757e-01	9.602e-01	HDAC9:183 HDAC4:242 CD86:253 CD40:389 DCAF1:400 FLT3:506
B Cell Activation Involved In Immune Res	0.03091495	12	7.108e-01	9.602e-01	MSH2:5881 GPR183:5881 CD40LG:5881 CD180:5881 LG4:5881 RNF8:5881
B Cell Chemotaxis (GO:0035754)	-0.04914117	5	7.036e-01	9.602e-01	GAS6:1685 PIK3CD:6785 HSD3B7:6785 CXCL13:6785 CYP7B1:6785 NA
B Cell Differentiation (GO:0030183)	0.04131844	44	3.435e-01	9.602e-01	HDAC9:183 HDAC4:242 DCAF1:400 FLT3:506 TPDS2:629 ZBTB7A:976
B Cell Mediated Immunity (GO:0019724)	-0.03123557	13	6.966e-01	9.602e-01	SLA2:1193 MFL2:1200 IL9R:1688 CD70:6785 MSH2:6785 FCGR2B:6785
B Cell Proliferation (GO:0042100)	0.08607651	15	2.485e-01	9.602e-01	CD40:389 MEFC2:979 IL10:5881 CD70:5881 IFNE:5881 PTPRC:5881
B Cell Receptor Signaling Pathway (GO:00)	0.09317267	26	1.003e-01	9.602e-01	CD38:283 CTLA4:625 VAV3:651 MEFC2:979 TEC:1073 SH2B2:1198
BMP Signaling Pathway (GO:0030509)	-0.06775707	42	1.291e-01	9.602e-01	SLC39A5:455 LEFTY1:659 RUNX2:710 GDF9:795 INHBE:813 INHBC:856
C–terminal Protein Amino Acid Modificati	-0.20084384	8	4.919e-02	9.602e-01	GPLD1:188 AGBL5:650 AGBL4:996 FOLH1:1730 AGTPBP1:6785 ATG16L1:6785
C–terminal Protein Deglutamylation (GO:0)	-0.30356961	4	3.550e-02	9.602e-01	AGBL5:650 AGBL4:996 FOLH1:1730 AGTPBP1:6785 NA NA
C–terminal Protein Lipidation (GO:0006050)	-0.14089015	3	3.980e-01	9.602e-01	GLPD1:188 ATG16L1:6785 ATG7:6785 NA NA NA
CD4–dicarboxylate Transport (GO:0015740)	0.03728866	10	6.833e-01	9.602e-01	SLC25A12:46 SLC25A18:569 SLC25A13:5881 SLC13A2:5881 SLC1A1:5881 UCP2:5881
CD4–positive, Alpha–Beta T Cell Activati	0.03089745	5	8.109e-01	9.602e-01	STOML2:5881 TNFSF4:5881 RSAD2:5881 NKG7:5881 HMGGB1:5881 NA
CD40 Signaling Pathway (GO:0020305)	0.03937031	7	7.183e-01	9.602e-01	CD86:253 CD40LG:5881 PHB2:5881 TNIP2:5881 ITGA5:5881 RNF31:5881
CENP–A Containing Chromatin Assembly (GO	-0.05491412	5	6.707e-01	9.602e-01	HJURP:1304 MIS18A:6785 CENPI:6785 OIP5:6785 NASP:6785 NA

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_IQ21_AMPLICON	-0.26593169	27	1.741e-06	1.128e-02	PBXIP1:32 ADAM15:55 RUSC1:113 MTX1:253 DCST2:335 THBS3:483
MIKKELSEN_ES_ICP_WITH_H3K4ME3	-0.05364367	532	2.757e-05	3.780e-02	SLC2A19:23 ZMYND8:23 PBXIP1:32 ZBTB4:22 ANXA3:48 WDCP:72
REACTOME_DNA_DOUBLE_STRAND_BREAK_REPAIR	-0.11504674	111	2.918e-05	3.780e-02	XRC5:31 ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419
REACTOME_MEIOTIC_RECOMBINATION	-0.22184976	31	1.926e-05	3.780e-02	ATM:60 MLH1:66 MLH3:90 BRCA1:206 RAD50:740 H2AX:792
REACTOME_DISEASES_OF_DNA_REPAIR	-0.17815757	47	2.413e-05	3.780e-02	ATM:60 MLH1:66 PMS2:169 BRCA1:206 MUTYH:211 HUS1:459
REACTOME_DEVELOPMENTAL_BIOLOGY	0.04592905	669	6.278e-05	5.816e-02	CACNA1S:17 TGFBI:22 CNTN1:26 CDK8:31 PPARGA:34 DEK:54
REACTOME_CYTOKINE_SIGNALING_IN_IMMUNE_SY	0.05420268	466	7.010e-05	5.816e-02	COL1A2:9 TGFBI:22 IL33:24 TNFRSF4:39 RELB:47 IL1RL2:121
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	-0.19802201	33	8.314e-05	5.816e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 RAD50:740
WP_DNA_REPAIR_PATHWAYS_FULL_NETWORK	-0.11467024	98	8.981e-05	5.816e-02	XRC5:31 ATM:60 MLH1:66 PMS2:169 BRCA1:206 MUTYH:211
WP_CILIOPATHIES	-0.09401272	148	8.217e-05	5.816e-02	DYNC2H1:5 KIAA0753:28 DDC2:192 SDCCAG8:200 DNAH11:217 CEP164:233
MIKKELSEN_NPC_ICP_WITH_H3K4ME3	-0.06393525	310	1.180e-04	6.371e-02	ZMYND8:23 PBXIP1:32 WDCP:72 KDM6B:180 ZNF608:216 CAVIN2:244
REACTOME_HDR_THROUGH_HOMOLOGOUS_RECOMBIN	-0.14677250	58	1.121e-04	6.371e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 HUS1:459
REACTOME_HDR_THROUGH_SINGLE_STRAND_ANNEA	-0.19290778	32	1.599e-04	7.504e-02	ATM:60 BRCA1:206 HUS1:459 RFC5:530 RAD50:740 DNA2:775
KIM_ALL_DISORDERS_CALB1_CORR_UP	0.06075307	330	1.622e-04	7.504e-02	SLC25A12:46 SLC20A1:49 NCKAP1:102 CHGB:117 ATP1A3:146 NPTN:192
REACTOME_INFECTIONOUS_DISEASE	0.04764474	539	1.789e-04	7.722e-02	TLR8:10 ENTDP1:18 TGFBI:22 ZH3GL3:38 AP1B1:44 TUBB4A:59
WP_22Q112_COPY_NUMBER_VARIATION_SYNDROME	0.12260096	77	2.032e-04	8.132e-02	HIRA:3 DGCR2:532 DEPDC5:60 ZDHHC8:130 P14KA:231 SERPIND1:250
KAUFFMANN_DNA_REPAIR_GENES	-0.08003677	181	2.135e-04	8.132e-02	XRC5:31 ATM:60 MLH1:66 MLH3:90 PMS2:169 BRCA1:206
REACTOME_HOMOLOGOUS_DIRECTED_REPAIR	-0.11198819	89	2.659e-04	9.566e-02	ATM:60 EME2:124 BRCA1:206 EME1:237 MUS81:419 HUS1:459
REACTOME_HOMOLOGOUS_DNA_PAIRING_AND_STRAI	-0.15910278	39	2.889e-04	1.909e-01	ATM:60 BRCA1:206 HUS1:459 RFC5:530 RAD50:740 DNA2:775
REACTOME_DNA_REPAIR	-0.06671598	226	5.768e-04	1.909e-01	USP45:10 XRC5:31 ATM:60 MLH1:66 EME2:124 PMS2:169
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.03884155	683	6.315e-04	1.947e-01	IWS1:7 COLLA2:9 TGFBI:22 SYMPK:23 CDK8:31 PPARGA:34
WP_INTERACTIONS_WITH_IMMUNE_CELLS_AND	0.20332916	23	7.390e-04	2.175e-01	TLR8:10 TGFBI:22 CD80:210 TGFBR2:234 CD86:253 CTLA4:625
WP_JOUBERT_SYNDROME	-0.12076775	64	8.438e-04	2.376e-01	ATM:60 CEP164:233 ARL13B:363 GNL3:148 RAB8A:681 TCTN2:693
REACTOME_RESOLUTION_OF_D_LOOP_STRUCTURES	-0.18559439	26	1.058e-03	2.854e-01	ATM:60 BRCA1:206 RAD50:740 DNA2:775 BRP1:1030 WRN:1040
WINTER_HYPOXIA_METAGENE	0.06982545	176	1.443e-03	3.738e-01	TGFBI:22 PPARGA:34 SLC20A1:49 ENG:51 LRP1:67 HK2:182
CONRAD_STEM_CELL	0.17512172	27	1.640e-03	4.000e-01	CD9:70 DNMT3B:140 MYC:187 PIWIL2:51 DKK3:305 RET:321
WP_T_CELL_MODULATION_IN_PANCREATIC_CANCE	0.15148390	36	1.669e-03	4.000e-01	ENTPD1:18 TGFBI:22 TNFRSF4:39 CD80:210 CD86:253 NT5E:365
WP_NETWORK_MAP_OF_SARSOCV2_SIGNALING_PAT	0.07281908	156	1.745e-03	4.035e-01	IL33:24 TNFSF10:78 RRM2:89 CD163:114 BID:143 ITIH3:171
REACTOME_ADAPTIVE_IMMUNE_SYSTEM	0.04240996	463	1.927e-03	4.160e-01	COL1A2:9 NFKBIE:32 HERC2:37 AP1B1:44 UBR2:50 TUBB4A:59
REACTOME_MEIOSIS	-0.12222087	54	1.907e-03	4.160e-01	ATM:60 MLH1:66 MLH3:90 BRCA1:206 STAG1:281 RAD50:740
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.06697460	180	2.003e-03	4.184e-01	TGFBI:22 TNFRSF4:39 TNFSF10:78 TGFBR2:234 TNFSF15:297 TNF:348
REACTOME_LOSS_OF_FUNCTION_OF_SMAD2_3_IN	0.39425241	5	2.265e-03	4.314e-01	TGFBI:22 ZFYVE9:86 TGFBR2:234 SMAD2:481 SMAD4:5881 NA
REACTOME_SIGNALING_BY_TGF_BETA_RECEPTOR_	0.39425241	5	2.265e-03	4.314e-01	TGFBI:22 ZFYVE9:86 TGFBR2:234 SMAD2:481 SMAD4:5881 NA
WP_SMC1SMC3_ROLE_IN_DNA_DAMAGE_CORNELIA_	-0.31272267	8	2.192e-03	4.314e-01	ATM:60 BRCA1:206 RAD50:740 PAXIP1:1061 MRE11:1398 RAD18:2001
REACTOME_SIGNALING_BY_INTERLEUKINS	0.05035864	312	2.349e-03	4.347e-01	COL1A2:9 TGFBI:22 IL33:24 IL1RL2:121 MYC:187 TRAF2:672
REACTOME_TRANSCRIPTIONAL_REGULATION_BY_T	0.17889236	24	2.423e-03	4.358e-01	DEK:54 YY1:160 MYC:187 MYBL2:396 TFAP2D:529 TFAP2C:202
PID_FANCONI_PATHWAY	-0.13271960	43	2.617e-03	4.581e-01	ATM:60 BRCA1:206 FANCD1:420 HUS1:459 RFC5:530 RAD50:740
REACTOME_RNA_POLYMERASE_II_TRANSCRIPTION	0.14659116	35	2.701e-03	4.603e-01	SYMPK:23 SLU7:139 THOCS198 LSM11:285 SLBP:409 CPSF1:443
MIKHAYLOVA_OXIDATIVE_STRESS_RESPONSE_VIA	0.42563137	4	3.196e-03	4.813e-01	OAT:694 CTSOD:761 AKR1B1:1136 PGAM1:1180 NA NA
KEGG_NON_HOMOLOGOUS_END_JOINING	-0.25695973	11	3.169e-03	4.813e-01	XRC5:31 RAD50:740 DNTT:1012 POLL:1047 XRC4:1239 NHEJ1:1351

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Hepatitis C	0.06750325	498	3.611e-07	3.526e-03	CLU:5 TLR8:10 TGFBI:22 PPARGA:34 ACE:36 DEPDC5:60
Encephalomyelitis	0.08513526	254	3.471e-06	1.695e-02	CLU:5 TLR8:10 ENTDP1:18 TGFBI:22 PPARGA:34 ACE:36
Dermatitis, Allergic Contact	0.13501732	83	2.202e-05	4.300e-02	PPARGA:34 ACE:36 CNTN3:179 CD80:210 MRC1:1224 HSD11B1:344
HIV Infections	0.05582680	511	2.045e-05	4.300e-02	TLR8:10 ENTDP1:18 TGFBI:22 GULIL:28 TRPV1:45 CD9:70
Psoriasis	0.05514324	522	2.134e-05	4.300e-02	TLR8:10 HCAR2:19 TGFBI:22 IL33:24 PPARGA:34 ACE:36
Aortic Aneurysm, Abdominal	0.08580122	196	3.780e-05	6.152e-02	COL1A2:9 TGFBI:22 PPARGA:34 ACE:36 FCGRT:63 LRP1:67
Seminoma	0.09992233	135	6.476e-05	9.035e-02	CLU:5 ACE:36 ENTDP6:99 AFP:118 DNMT3B:140 GTTLC3:177
Dermatitis	0.09661719	138	9.363e-05	1.046e-01	TGFBI:22 IL33:24 PPARGA:34 TRPA1:40 CLK7:74 LPIN2:110
Inflammatory dermatosis	0.10557842	115	9.636e-05	1.046e-01	TGFBI:22 IL33:24 PPARGA:34 TRPA1:40 LPIN2:110 PTGDS:164
Myasthenia Gravis	0.10245538	120	1.112e-04	1.086e-01	TGFBI:22 TNFRSF4:39 DNMT3B:140 CD86:253 ITGAX:274 KRT5:277
Scleroderma	0.10185243	119	1.302e-04	1.156e-01	COL1A2:9 TGFBI:22 ENG:51 RTCA:94 YY1:160 COMP:169
Dermatitis, Atopic	0.06237939	316	1.563e-04	1.174e-01	TGFBI:22 IL33:24 PPARGA:34 RELB:47 KLF5:58 FCGRT:63
Eczema	0.06337804	306	1.556e-04	1.174e-01	TGFBI:22 IL33:24 PPARGA:34 KLF5:58 KLF7:74 TRPM1:87
Malignant neoplasm of stomach	0.03067075	1450	1.801e-04	1.256e-01	CLU:5 COL1A2:9 TGFBI:22 SYMPK:23 IL33:24 CNTN1:26
Lymphoma, T–Cell, Cutaneous	0.08092866	177	2.182e-04	1.421e-01	TGFBI:22 TNFSF10:78 SCPEP1:128 HDAC9:183 MYC:187 TRAF2:202
Hepatocarcinogenesis	0.05865038	326	3.077e-04	1.878e-01	CLU:5 TGFBI:22 GULIL:28 PPARGA:34 TRIM24:57 TNFSF10:78
Epilepsy, Temporal Lobe	0.09330872	122	3.874e-04	2.033e-01	GULIL:28 ACE:36 TRPV1:45 DEPDC5:60 TNFSF10:78 SCN2A:145
Chronic Lymphocytic Leukemia	0.03985666	708	3.954e-04	2.033e-01	ENTPD1:18 TGFBI:22 NFKBIE:32 PPARGA:34 ACE:36 AQP1:42
Pemphigus Vulgaris	0.15896573	42	3.692e-04	2.033e-01	FCGRT:63 GJA8:120 DSG3:213 TNF:348 CD28:416 TNFRSF9:536
Ulcerative Colitis	0.04393224	585	4.169e-04	2.036e-01	CLU:5 TGFBI:22 IL33:24 PPARGA:34 ACE:36 HERC2:37
Cataract, Pulverulent	0.47890938	5	5.237e-04	2.325e-01	GJA8:120 GJA3:238 CYPBB1:741 HSF:4954 BFSP2:1137 NA
Giant Cell Glioblastoma	0.12864985	61	5.204e-04	2.325e-01	CD9:70 TNFSF10:78 MYC:187 NOTCH3:333 BHLHE40:403 FN1:425
Tuberculosis, Pulmonary	0.09154807	117	6.489e-04	2.391e-01	TLR8:10 TGFBI:22 TRPV5:25 PDE6B:30 INTS4:69 CD9:70
Cold intolerance	0.40614217	6	5.710e-04	2.391e-01	PPARGA:34 TRPA1:40 TRPV1:45 UCP1:1497 IRF4:560 LPIN1:5881
Crohn Disease	0.04203590	573	7.100e-04	2.391e-01	TLR8:10 ENTDP